

Barcode clustering benchmark

Preserved v14 + faster startup | completed Shepherd sensitivity comparison

60 independent libraries per simulated design. FN / FP: mean counts. F1: mean % (SD). Time: median workflow seconds. Milo: one fixed reference.

Method	FN (lower)	FP (lower)	F1 % (higher)	Time s (lower)
Random N20: substitutions + repeat-dependent indels				
barbac Hamming	124.92	18.30	99.28004 (0.0758)	1.40
barbac LV + Poisson	124.90	18.25	99.28038 (0.0752)	1.76
Shepherd*	126.53	19.83	99.26421 (0.0766)	4.67
Starcode sphere	129.72	22.32	99.23569 (0.0797)	3.29
Starcode MP	118.18	142.98	98.69582 (0.1112)	3.22
Bartender	125.17	48.03	99.13063 (0.0921)	1.89
Anchored N20 + AA/TT: substitutions + repeat-dependent indels				
barbac Hamming	127.70	24.90	99.23302 (0.0736)	1.52
barbac LV + Poisson	127.67	24.08	99.23726 (0.0741)	2.05
Shepherd*	129.65	25.05	99.22239 (0.0796)	13.45
Starcode sphere	132.63	27.65	99.19431 (0.0829)	3.77
Starcode MP	119.95	191.83	98.44674 (0.1384)	3.67
Bartender	127.73	137.97	98.67218 (0.0786)	2.31
Milo / Johnson published reference simulation				
barbac Hamming	469	87	99.72147	17.50
barbac LV + Poisson	471	83	99.72246	33.65
Shepherd*	470	85	99.72196	165.23
Starcode sphere	771	353	99.43682	170.68
Starcode MP	532	560	99.45408	193.39
Bartender	496	795	99.35546	41.77

Bold/shading marks numerical leaders, not statistical significance. All other methods retain their frozen outputs and accepted timing observations. No Barbac code or simulation data changed.

*Shepherd supplementary configuration: distance 3, author-documented Bayes threshold -4; known generating substitution rate 0.004 on every new simulation. Milo retains automatic rate estimation. The original configuration used threshold +4 and automatic rates, failing on 10/120 new inputs. Its results and PDF remain preserved separately.

This configuration was registered after the original outcomes were available. It is an explicitly post hoc sensitivity comparison, not the original confirmatory result. Supplying Shepherd the generating substitution rate favours a correctly specified model; no truth identities or repeat-specific indel rates are supplied. This is not an all-default run.

New simulations: 10,000 true identities and one million expected reads per library; paired designs share variable identities and parent counts. Substitutions: 0.4% per base. Indels: unscaled archived homopolymer-specific rates. For repeats beyond calibrated length 13, further indel recursion stops while reads and subsequent substitutions are retained (249 affected reads among 119,992,068; 16/120 libraries).

FN counts true identities absent from inferred centroids; FP counts inferred identities absent from truth. F1 includes zero-read true identities. Positive-read F1, read assignment accuracy and abundance metrics are in the companion results. The fixed Milo simulation contains 100,000 identities and 24,996,128 reads and was used during development; it cannot establish population-level significance.

Timing includes fresh-worker/tool startup, required conversion and canonical exports; common staging and scoring are excluded. Shepherd times come from a later session and are descriptive, not a randomized paired-session speed comparison. Lightweight diagnostic file reads overlapped early calls. Original methods retain the previously documented sleep-repair observations.

Other settings: distance 3, one thread. Barbac support ordering, ratio 20, configured proxy 0.005, design scoring off; LV Poisson on, Hamming rare-indel rescue retained. Starcode MP ratio 5. Bartender seed length 5, step 1, z=5, cutoff 1. Full protocol, preserved failures and supplementary paired uncertainty: README.md and RESULTS.md.